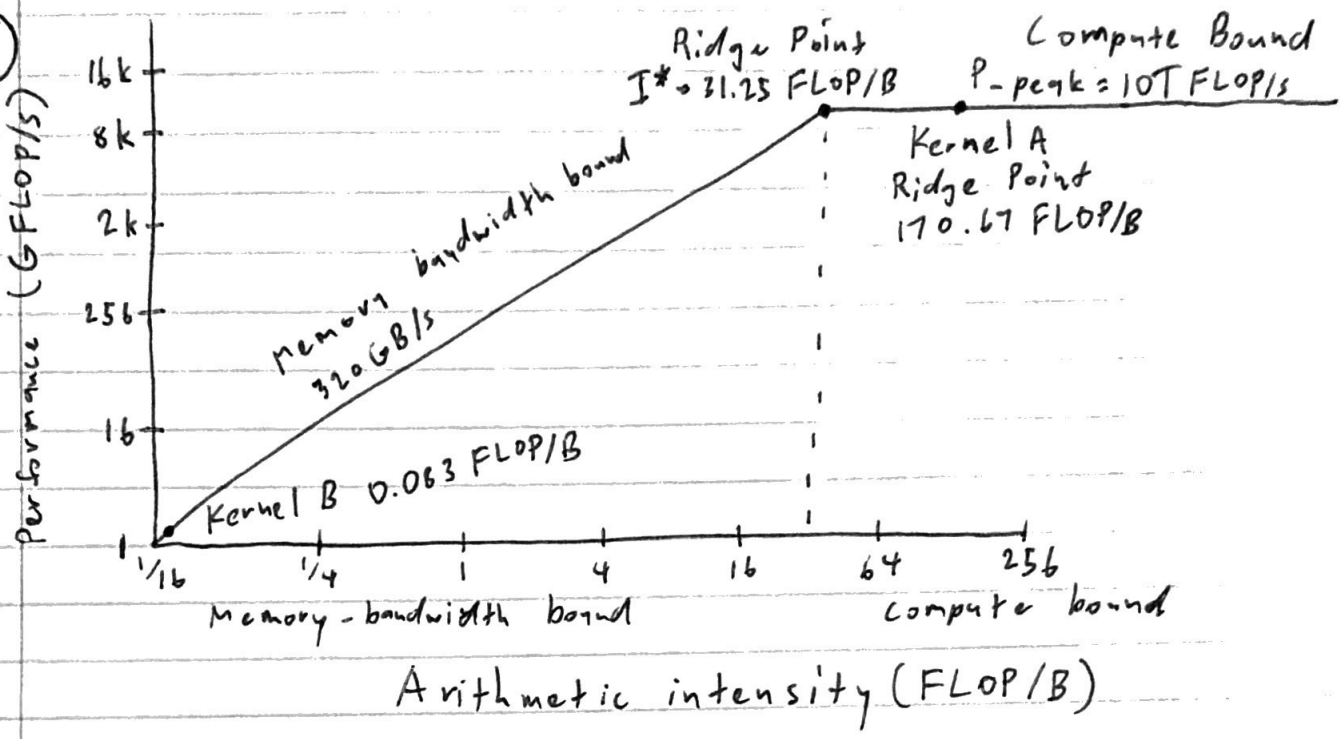
1



2 $2 \times 1024^3 = 2{,}147{,}483{,}648$ FLOPs ≈ 2.15 GFLOPs

$3 \times 1024^2 = 3145728 \times 4 = 12582912$ bytes transfered

2.15 GFLOPs / 12582912 bytes trans = 170.67 FLOP/B

3 4,194,304 FLOPs

$3 \times 4{,}194{,}304 \times 4 = 50{,}331{,}648$ bytes transferred

$4194304 / 50331648 \approx 0.083$ FLOP/B

4 a) Compute bound for Kernel A, Memory-bandwidth bound for Kernel B.

b) It would by 10TFLOP/s since that is the roof

c) Increasing DRAM bandwidth, which would lower ridge point